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Course: CSCI 5499B

Project: Project 1 – Underwater

Game Overview / Concept

My game is a 2D underwater platformer where the player controls a worm that has fallen off of a fishing hook. The main objective is to make it through the underwater environment while avoiding fish and other obstacles. The player can use seaweed to hide from fish and moving hooks as platforms to get across parts of the level. The player completes the level by reaching the safe spot at the end. After completing both levels, the player wins the game.

Game Background / Setting

The game takes place underwater. The player is a worm that has fallen off of a fishing hook and now has to make their way through the underwater environment without being eaten by fish. The worm's goal is to reach the sandcastle at the end, which acts as a safe spot. Along the way, the worm can hide in seaweed to avoid fish and use fishing hooks as moving platforms.

Levels

Level 1 -

Level 1 is designed to introduce the player to the main mechanics of the game. The player first learns how to move and jump. As they continue through the level, they encounter fish and learn that touching a fish causes them to die. Seaweed is then used to teach the player that they can hide from the fish instead of always trying to avoid them. The player also encounters moving fishing hooks that act as platforms. These teach the player how to time their jumps based on the movement of the platform. By the end of Level 1, the player should understand movement, jumping, avoiding fish, hiding in seaweed, using moving platforms, and how to reach the end of a level.

Level 2 -

Level 2 is designed to make the player apply the mechanics learned in Level 1 in more difficult situations. Platforms and enemies are used together so that the player has to pay attention to multiple things at once. The player also has to use more strategy to decide when it is safe to move, when they should hide, and when

they should jump onto a moving platform. This is especially important for your assignment because your professor specifically wants one level to teach the concepts and the next level to have the player apply those concepts in new ways.

Core Mechanics & Controls

- Movement
 - Aswd for up, down, left and right
 - Arrow keys for up, down, left and right
 - Spacebar can also be used for jump
- Fish
 - Will move back and forth
 - If player touches fish they die
- Seaweed
 - Hides player and makes it so fish's collision box won't trigger death
- Moving hook/platform
 - Move up and down at different paces and serve as a platform
- Death
 - Player touches fish
 - Falling too far off screen
 - Death animation/sinking
 - Given Game Over screen and Try Again button
- Level completion
 - Level 1 end zone goes to Level 2
 - Level 2 end zone goes to Win screen

Game Rules

1. The player must reach the safe area at the end of each level to progress.
2. The player dies when caught by a fish unless they are hiding in seaweed.
3. Falling too far below the level causes the player to die.
4. The player can jump onto stationary platforms and moving fishing hooks.
5. Seaweed acts as a safe area from fish.
6. After dying, the player can use the Try Again button to restart the current level.
7. Completing Level 1 advances the player to Level 2.
8. Completing Level 2 wins the game.

User Interface

- Start Screen

- Level 2: Displays "level 2" once loaded up
- Game Over: Gives a retry button
- Win Screen: Option to restart or quit game

Playtesting and Iterations

Iterations:

- Iteration 1 – basic player movement and animation
- Iteration 2 – fish enemies and player death
- Iteration 3 – seaweed hiding mechanic
- Iteration 4 – moving hook platforms
- Iteration 5 – game over and restart system
- Iteration 6 – start screen and level progression
- Iteration 6.5 – added 2d camera to follow player
- Iteration 7 – level 2 and increased difficulty
- Iteration 8 – level 2 fish collision fix
- Iteration 9 – falling-off-screen death fix
- Iteration 10 – win screen and restart buttons fixed
- Iteration 11 – tweaking levels difficulty and changing tilemaps